PRACTICE SHEET

1. What is the area under the curve $y = |x| + |x - 1|$ between $x = 0$ and $x = 1$?
(a) 1/2 (b) 1
(c) 3/2 (d) 2

2. What is the area enclosed by the curve $2x^2 + y^2 = 1$?
(a) 2π (b) π
(c) $\pi/2$ (d) $\pi/\sqrt{2}$

3. What is the area of the region bounded by the line $3x - 5y = 15$, $x = 1$, $x = 3$ and x – axis in sq unit?
(a) 36/5 (b) 18/5
(c) 9/5 (d) 3/5

4. What is the area of he ellipse $4x^2 + 9y^2 = 1$
(a) 6π (b) $\pi/36$
(c) $\pi/6$ (d) $\pi/\sqrt{6}$

5. What is the value of k if the area bounded by the curve $y = \sin kx$, $y = 0$, $x = \pi / (3k)$ is 3 sq unit?
(a) 1/2 (b) 1
(c) 3/2 (d) 2

6. What is the area bounded by the curve $y = 4x - x^2 - 3$ and the x – axis?
(a) 2/3 sq unit (b) 4/3 sq unit
(c) 5/3 sq unit (d) 4/5 sq unit

7. What is the area under the curve $f(x) = xe^x$ above the X-axis and between the lines $x = 0$ and $x = 1$?
(a) 1/2 sq. unit (b) 1 sq. unit
(c) 3/2 sq. units (d) 2 sq. units

8. If $f(x) = 1 - \frac{x^2}{4}$, $x \in [-2,2]$, then find the area covered by X-axis.
(a) 8/3 sq. units (b) 4/3 sq. units
(c) 2/3 sq. unit (d) 1/3 sq. unit

9. What is the area bounded by the curve $y = x^2$ and the line $y = 16$?
(a) 32/3 (b) 64/3
(c) 256/3 (d) 128/3

10. What is the area enclosed between the curve $y^2 = 12x$ and the lines $x = 0$ and $y = 6$?
(a) 2 sq. units (b) 4 sq. units
(c) 6 sq. units (d) 8 sq. units

ANSWER KEYS																			
1.	b	**2.**	d	**3.**	b	**4.**	c	**5.**	a	**6.**	b	**7.**	d	**8.**	a	**9.**	c	**10.**	c